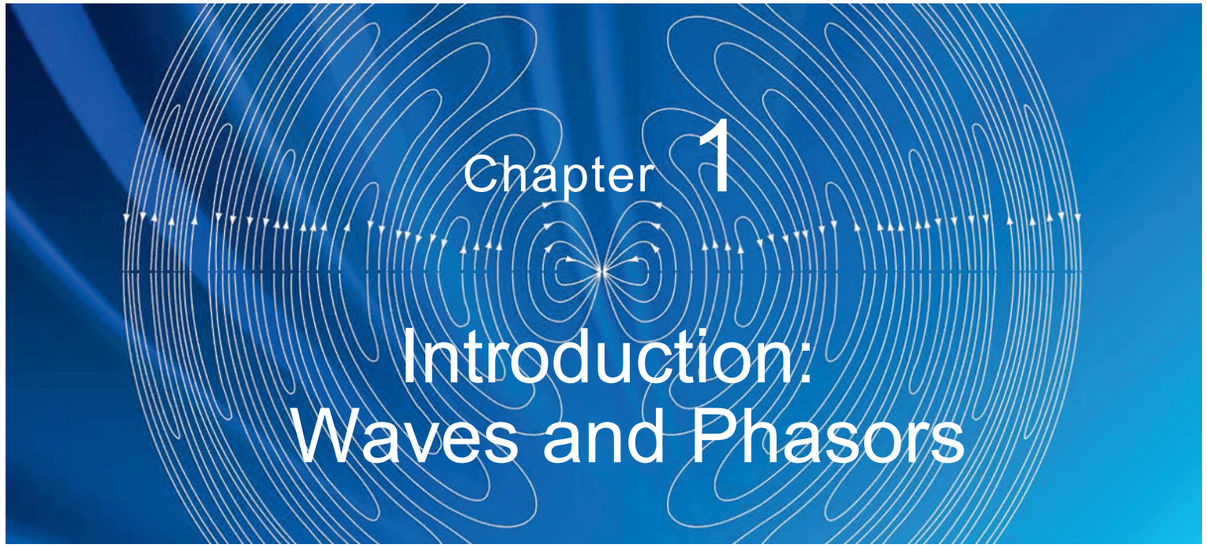


Chapter 1
Introduction: Waves and Phasors

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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Describe the basic properties of electric and magnetic forces.
2. Ascribe mathematical formulations to sinusoidal waves traveling in both lossless and lossy media.
3. Apply complex algebra in rectangular and polar forms.
4. Apply the phasor-domain technique to analyze circuits driven by sinusoidal sources.

绪论：波与相量

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学习目标

学习本章后，你应能够：

1. 描述电力与磁力的基本性质。
2. 用数学形式描述正弦波在无损和有损介质中的传播。
3. 运用直角坐标形式和极坐标形式的复数代数。
4. 应用相量域方法，分析由正弦信号源激励的电路。

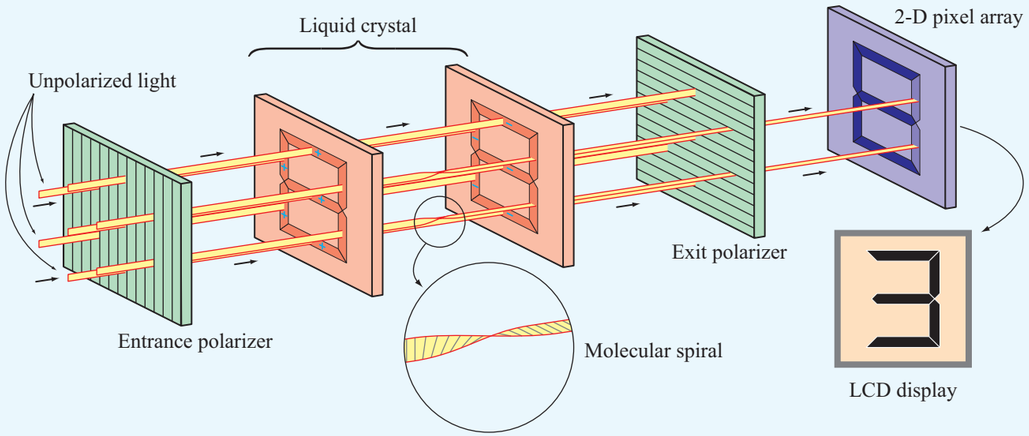


Figure 1-1 2-D LCD array.

Overview

Liquid crystal displays (***LCDs***) have become integral parts of many electronic consumer products, ranging from alarm clocks and cell phones to laptop computers and television systems. LCD technology relies on special electrical and optical properties of a class of materials known as ***liquid crystals***, which are neither pure solids nor pure liquids but rather a hybrid of both. The molecular structure of these materials is such that when light travels through them, the polarization of the emerging light depends on whether or not a voltage exists across the material. Consequently, when no voltage is applied, the exit surface appears bright. Conversely, when a voltage of a certain level is applied across the LCD material, no light passes through it, resulting in a dark pixel. In-between voltages translate into a range of grey levels. By controlling the voltages across individual pixels in a two-dimensional array, a complete image can be displayed (**Fig. 1-1**). Color displays are composed of three subpixels with red, green, and blue filters.

► The polarization behavior of light in a LCD is a prime example of how electromagnetics is at the heart of electrical and computer engineering. ◄

The subject of this book is applied ***electromagnetics*** (***EM***), which encompasses the study of both static and dynamic electric and magnetic phenomena and their engineering applications. Primary emphasis is placed on the fundamental properties of dynamic (time-varying) electromagnetic fields because of their greater relevance to practical applications, including wireless and optical communications, radar, bioelectromagnetics, and high-speed microelectronics. We study wave propagation in guided media, such as coaxial transmission lines, optical fibers, and waveguides; wave reflection and transmission at interfaces between dissimilar media; radiation by antennas, and several other related topics. The concluding chapter is intended to illustrate a few aspects of applied EM through an examination of design considerations associated with the use and operation of radar sensors and satellite communication systems.

We begin this chapter with a chronology of the history of electricity and magnetism. Next, we introduce the fundamental electric and magnetic field quantities of electromagnetics, as well as their relationships to each other and to the electric charges and currents that generate them. These relationships constitute the underpinnings of the study of electromagnetic phenomena. Then, in preparation for the material presented in Chapter 2, we provide short reviews of three topics: traveling waves, complex numbers, and phasors, which are all useful in solving time-harmonic problems.

概述

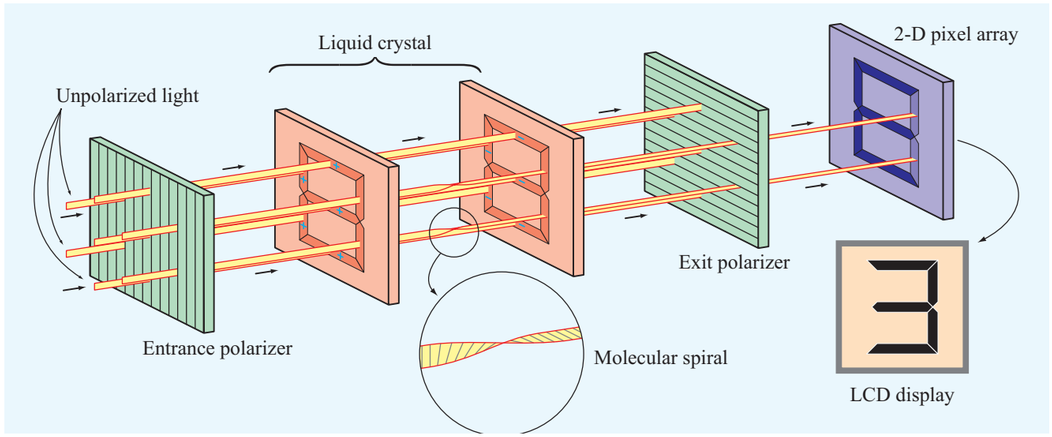


图 1-1 二维液晶显示阵列。

图中标注：非偏振光；入射偏振片；液晶；分子螺旋；出射偏振片；二维像素阵列；LCD 显示。

液晶显示器（LCD）已成为许多消费电子产品的重要组成部分，从闹钟、手机到笔记本电脑和电视系统，均可见其应用。液晶技术利用一类称为液晶的材料所具有的特殊电学和光学性质。这类材料既不是纯粹的固体，也不是纯粹的液体，而兼有两者的某些特点。其分子结构使得光穿过材料后，出射光的偏振状态取决于材料两端是否施加电压。

因此，未施加电压时，出射面呈亮态；施加一定电压后，光无法透过，像素呈暗态。介于两者之间的电压对应不同灰度。通过控制二维阵列中各像素的电压，就能显示完整图像，如图 1-1 所示。彩色显示的每个像素由分别带有红、绿、蓝滤光片的三个子像素组成。

> **液晶中光的偏振行为，是电磁学处于电气与计算机工程核心地位的一个典型例子。**

本书研究应用电磁学（EM），内容包括静态与动态电、磁现象及其工程应用。重点讨论动态（时变）电磁场的基本性质，因为它们与无线和光通信、雷达、生物电磁学、高速微电子等实际应用关系更为密切。我们将学习同轴传输线、光纤和波导等导波介质中的波传播，不同介质分界面上的反射与透射，天线辐射及相关内容。最后一章通过雷达传感器和卫星通信系统的设计与运行，展示应用电磁学的若干实例。

本章先简述电与磁的发展历史，再介绍基本电磁场量，以及这些场量彼此之间、与产生它们的电荷和电流之间的关系。这些关系构成研究电磁现象的基础。随后，为第 2 章作准备，我们简要复习行波、复数与相量，这三项内容都适用于求解时谐问题。

1-1 Historical Timeline

The history of EM may be divided into two overlapping eras. The first is the ***classical era***, during which the fundamental laws of electricity and magnetism were discovered and formulated. Building on these formulations, the ***modern era*** of the past 100 years ushered in the birth of the field of applied EM as we know it today.

1-1.1 EM in the Classical Era

Chronology 1-1 provides a timeline for the development of electromagnetic theory in the classical era. It highlights those discoveries and inventions that have impacted the historical development of EM in a very significant way, even though the selected discoveries represent only a small fraction of those responsible for our current understanding of electromagnetics. As we proceed through this book, some of the names highlighted in Chronology 1-1, such as those of Coulomb and Faraday, will appear again as we discuss the laws and formulations named after them.

The attractive force of magnetite was reported by the Greeks some 2800 years ago. It was also a Greek, ***Thales of Miletus***, who first wrote about what we now call static electricity: He described how rubbing amber caused it to develop a force that could pick up light objects such as feathers. The term "*electric*" first appeared in print around 1600 in a treatise on the (electric) force generated by friction, authored by the physician to Queen Elizabeth I, ***William Gilbert***.

About a century later, in 1733, ***Charles-François du Fay*** introduced the notion that electricity involves two types of "fluids," one "positive" and the other "negative," and that like-fluids repel and opposite-fluids attract. His notion of a fluid is what we today call electric charge. The invention of the capacitor in 1745, originally called the ***Leyden jar***, made it possible to store significant amounts of electric charge in a single device. A few years later, in 1752, ***Benjamin Franklin*** demonstrated that lightning is a form of electricity. He transferred electric charge from a cloud to a Leyden jar via a silk kite flown in a thunderstorm. The collective eighteenth-century knowledge about electricity was integrated in 1785 by ***Charles-Augustin de Coulomb***, in the form of a mathematical formulation characterizing the electrical force between two charges in terms of their strengths and polarities and the distance between them.

The year 1800 is noted for the development of the first electric battery by ***Alessandro Volta***, and 1820 was a banner year for discoveries about how electric currents induce magnetism. This knowledge was put to good use by ***Joseph Henry***, who developed one of the earliest electromagnets and dc (direct current) electric motors. Shortly thereafter, ***Michael Faraday*** built the first electric generator (the converse of the electric motor). Faraday, in essence, demonstrated that a changing magnetic field induces an electric field (and hence a voltage). The converse relation, namely that a changing electric field induces a magnetic field, was first proposed by ***James Clerk Maxwell*** in 1864 and then incorporated into his four (now) famous equations in 1873.

► Maxwell's equations represent the foundation of classical electromagnetic theory. ◄

Maxwell's theory, which predicted the existence of electromagnetic waves, was not fully accepted by the scientific community at that time. It was later verified experimentally by means of radio waves by ***Heinrich Hertz*** in the 1880s. X-rays, another member of the EM family, were discovered in 1895 by ***Wilhelm Röntgen***. In the same decade, ***Nikola Tesla*** was the first to develop the ac motor, which was considered a major advance over its predecessor, the dc motor.

Despite the advances made in the 19th century in our understanding of electricity and magnetism and how to put them to practical use, it was not until 1897 that the fundamental carrier of electric charge, the electron, was identified and its properties quantified by ***Joseph Thomson***. The ability to eject electrons from a material by shining electromagnetic energy, such as light, on it is known as the ***photoelectric effect***.

► To explain the photoelectric effect, ***Albert Einstein*** adopted the quantum concept of energy that had been advanced a few years earlier (1900) by ***Max Planck***. Symbolically, this step represents the bridge between the classical and modern eras of electromagnetics. ◄

1-1.2 EM in the Modern Era

Electromagnetics play a role in the design and operation of every conceivable electronic device, including the diode, transistor, integrated circuit, laser, display screen, bar-code reader, cell phone, and microwave oven, to name but a few. Given the breadth and diversity of these applications (**Fig. 1-2**), it is far more difficult to construct a meaningful timeline for the modern era than for the classical era. That said, one can develop timelines for specific technologies and link their milestone innovations to EM. Chronologies 1-2 and 1-3 present timelines for the development of telecommunications and computers, technologies that have become integral parts of today's societal infrastructure. Some of the entries in these chronologies refer to specific inventions, such as the telegraph, the transistor, and the laser. The operational principles and capabilities of some of these technologies are highlighted in

HISTORICAL TIMELINE

1-1 历史沿革

电磁学的历史可以分为两个相互交叠的时期。第一个是经典时期，电与磁的基本规律在这一时期被发现并形成数学表述。在此基础上，近百年的现代时期孕育了我们今天所认识的应用电磁学。

1-1.1 经典时期的电磁学

年表 1-1 概述了经典电磁理论的发展，重点列出对其历史进程产生重大影响的发现与发明。这些条目只是构成当今电磁学认识的诸多成果中的一小部分。书中介绍有关定律和公式时，库仑、法拉第等人物的名字还会再次出现。

约 2800 年前，希腊人已记录磁铁矿的吸引力。米利都的泰勒斯最早记述了今天所说的静电现象：琥珀经摩擦后能吸起羽毛等轻小物体。约 1600 年，英国女王伊丽莎白一世的御医威廉·吉尔伯特在讨论摩擦所产生电力的著作中，首次在印刷文献中使用“electric”一词。

1733 年，夏尔-弗朗索瓦·杜费提出电涉及“正”“负”两种流体，同种相斥、异种相吸；这种“流体”就是今天所说的电荷。1745 年发明的电容器最初称为莱顿瓶，使人们能在单个器件中储存较多电荷。1752 年，本杰明·富兰克林通过在雷雨中放飞丝绸风筝，将云中的电荷引入莱顿瓶，证明闪电是一种电现象。1785 年，夏尔-奥古斯丁·库仑将十八世纪的有关认识归纳为数学关系，用电荷量、极性及间距描述两个电荷之间的电力。

1800 年，亚历山德罗·伏打制成第一种电池。1820 年，人们在电流产生磁效应方面取得重要发现。约瑟夫·亨利据此研制出早期电磁铁和直流电动机。随后，迈克尔·法拉第制造出第一台发电机，其作用与电动机相反；他证明变化的磁场会产生电场，从而产生电压。詹姆斯·克拉克·麦克斯韦于 1864 年提出相反的关系，即变化的电场会产生磁场，并于 1873 年将其纳入著名的四个方程。

麦克斯韦方程组是经典电磁理论的基础。

麦克斯韦理论预言了电磁波的存在，但当时尚未被科学界充分接受。十九世纪八十年代，海因里希·赫兹通过无线电波实验验证了这一理论。1895 年，威廉·伦琴发现了电磁波家族的另一成员：X 射线。同一年代，尼古拉·特斯拉率先研制交流电动机，这被认为是相对直流电动机的重要进步。

尽管十九世纪对电与磁的理解及应用已取得重大进展，直到 1897 年，约瑟夫·汤姆孙才确认电子这一基本电荷载体，并定量研究其性质。用光等电磁能量照射材料而使电子逸出的现象，称为光电效应。

为解释光电效应，阿尔伯特•爱因斯坦采用了马克斯•普朗克于 1900 年提出的能量量子概念。象征性地说，这一步连接了电磁学的经典时期与现代时期。

1-1.2 现代时期的电磁学

电磁学参与几乎所有电子器件的设计与运行，例如二极管、晶体管、集成电路、激光器、显示屏、条形码阅读器、手机和微波炉。由于应用广泛且多样（见图 1-2），为现代时期编写一份有代表性的总年表比经典时期困难得多。不过，可以为具体技术编写年表，并将关键创新与电磁学联系起来。年表 1-2 和 1-3 分别介绍电信和计算机的发展；它们已成为现代社会基础设施的重要组成部分。条目中包含电报、晶体管、激光器等具体发明。本书穿插的“技术简介”专栏将介绍其中部分技术的工作原理和性能。

编排说明：末段最后一句接原书第 11 页开头，中文在此合并为完整句。

Chronology 1-1: TIMELINE FOR ELECTROMAGNETICS IN THE CLASSICAL ERA

Electromagnetics in the Classical Era

ca. 900 BC — Legend has it that, while walking across a field in northern Greece, a shepherd named Magnus experiences a pull on the iron nails in his sandals by the black rock he is standing on. The region was later named Magnesia and the rock became known as magnetite [a form of iron with permanent magnetism].

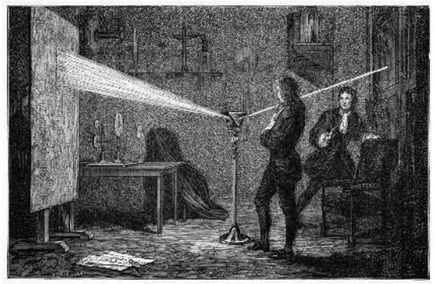
ca. 600 BC — Greek philosopher **Thales** describes how amber, after being rubbed with cat fur, can pick up feathers [static electricity].



ca. 1000 — Magnetic compass used as a navigational device.

1600 — **William Gilbert** (English) coins the term electric after the Greek word for amber (elektron), and observes that a compass needle points north–south because the Earth acts as a bar magnet.

1671 — **Isaac Newton** (English) demonstrates that white light is a mixture of all the colors.



1733 — **Charles-François du Fay** (French) discovers that electric charges are of two forms and that like charges repel and unlike charges attract.

1745 — **Pieter van Musschenbroek** (Dutch) invents the Leyden jar, which was the first electrical capacitor.

1752 — **Benjamin Franklin** (American) invents the lightning rod and demonstrates that lightning is electricity.



1785 — **Charles-Augustin de Coulomb** (French) demonstrates that the electrical force between charges is proportional to the inverse of the square of the distance between them.

1800 — **Alessandro Volta** (Italian) develops the first electric battery.



1820 — **Hans Christian Oersted** (Danish) demonstrates the interconnection between electricity and magnetism through his discovery that an electric current in a wire causes a compass needle to orient itself perpendicular to the wire.



1820 — **André-Marie Ampère** (French) notes that parallel currents in wires attract each other and opposite currents repel.



1820 — **Jean-Baptiste Biot** (French) and **Félix Savart** (French) develop the Biot–Savart law relating the magnetic field induced by a wire segment to the current flowing through it.

4 CHAPTER 1 INTRODUCTION: WAVES AND PHASORS

年表 1-1：古典时代电磁学的时间表

古典时代的电磁学

约公元前900年 传说牧羊人马格努斯在希腊北部行走时，脚下黑色岩石吸住了凉鞋上的铁钉。该地区后来称为Magnesia，这种具有天然磁性的岩石称为磁铁矿。

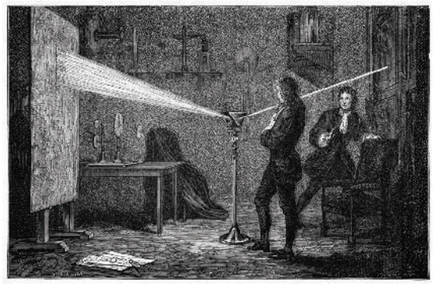
约公元前600年 希腊哲学家泰勒斯记述：琥珀经猫毛摩擦后能够吸起羽毛（静电现象）。



约 1000 用作导航设备的磁罗盘。

1600 年威廉·吉尔伯特（英国）根据希腊语中的琥珀（elektron）创造了“电”一词，并观察到罗盘指针指向南北，因为地球充当了条形磁铁。

1671年 英国科学家艾萨克·牛顿证明，白光是各种颜色光的混合。



1733 年 Charles-François du Fay（法国）发现电荷有两种形式，同性电荷相斥，异性电荷相吸。

1745 Pieter van Musschenbroek（荷兰人）发明了莱顿瓶，这是第一个电容器。

1752年本杰明·富兰克林（美国人）发明了避雷针，并证明闪电就是电。



1785 Charles-Augustin de Coulomb（法国）证明，电荷之间的电力与它们之间的距离的平方成反比。

1800 亚历山德罗·沃尔塔（意大利人）开发出第一块电池。



1820 汉斯·克里斯蒂安·奥斯特（丹麦人）通过发现电线中的电流使罗盘针垂直于电线定向，证明了电和磁之间的相互联系。



1820 年 André-Marie Ampère（法国）指出，电线中的平行电流相互吸引，相反的电流相互排斥。



1820 年 Jean-Baptiste Biot（法国）和 Félix Savart（法国）制定了 Biot-Savart 定律，将导线段感应的磁场与流过导线的电流联系起来。

Chronology 1-1: TIMELINE FOR ELECTROMAGNETICS IN THE CLASSICAL ERA (continued)

Electromagnetics in the Classical Era

1827 **Georg Simon Ohm** (German) formulates Ohm's law relating electric potential to current and resistance.

1827 **Joseph Henry** (American) introduces the concept of inductance and builds one of the earliest electric motors. He also assisted Samual Morse in the development of the telegraph.

1831 **Michael Faraday** (English) discovers that a changing magnetic flux can induce an electromotive force.



1835 **Carl Friedrich Gauss** (German) formulates Gauss's law relating the electric flux flowing through an enclosed surface to the enclosed electric charge.

Gauss' Law for Electricity

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{inside}}{\varepsilon_0}$$

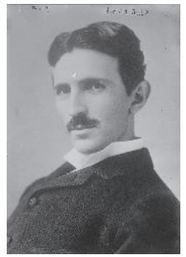
1873 **James Clerk Maxwell** (Scottish) publishes his *Treatise on Electricity and Magnetism*, in which he unites the discoveries of Coulomb, Oersted, Ampère, Faraday, and others into four elegantly constructed mathematical equations, now known as Maxwell's Equations.



1887 **Heinrich Hertz** (German) builds a system that can generate electromagnetic waves (at radio frequencies) and detect them.



1888 **Nikola Tesla** (American) invents the ac (alternating current) electric motor.

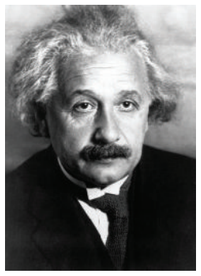


1895 **Wilhelm Röntgen** (German) discovers X-rays. One of his first X-ray images was of the bones in his wife's hands. [1901 Nobel prize in physics.]



1897 **Joseph John Thomson** (English) discovers the electron and measures its charge-to-mass ratio. [1906 Nobel prize in physics.]

1905 **Albert Einstein** (German-American) explains the photoelectric effect discovered earlier by Hertz in 1887. [1921 Nobel prize in physics.]



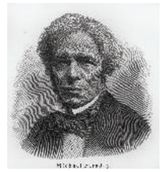
年表 1-1：古典时代电磁学的时间表（续）

古典时代的电磁学

1827 Georg Simon Ohm（德国）制定了将电势与电流和电阻联系起来的欧姆定律。

1827 Joseph Henry（美国人）引入了电感的概念并制造了最早的电动机之一。
他还协助塞缪尔·莫尔斯开发电报。

1831 年迈克尔·法拉第（英语）发现变化的磁通量可以感应出电动势。



1835年 卡尔·弗里德里希·高斯（德国）制定了高斯定律，将流过封闭表面的电通量与封闭的电荷联系起来。

Gauss' Law for Electricity

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{inside}}{\varepsilon_0}$$

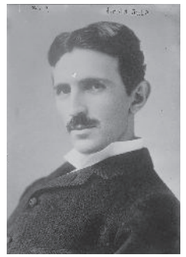
分段阅读_第 1873 章
（苏格兰）出版了他的《电和磁学论文》，其中他将库仑、奥斯特、安培、法拉第等人的发现统一到四个精心构造的数学方程中，现在称为麦克斯韦方程组。



1887年 海因里希·赫兹（德国）建立了一个可以产生电磁波（射频）并检测它们的系统。



1888年 尼古拉·特斯拉（美国人）发明了交流电动机。



1895 年 Wilhelm Röntgen（德国）发现 X 射线。 他的第一张 X 射线图像是他妻子手中的骨头。 [1901 年诺贝尔物理学奖。]



1897年 约瑟夫·约翰·汤姆森（英语）发现了电子并测量了其荷质比。 [1906年诺贝尔物理学奖。]

1905年 阿尔伯特·爱因斯坦（德裔美国人）解释了赫兹于 1887 年发现的光电效应。[1921 年诺贝尔物理学奖。]

